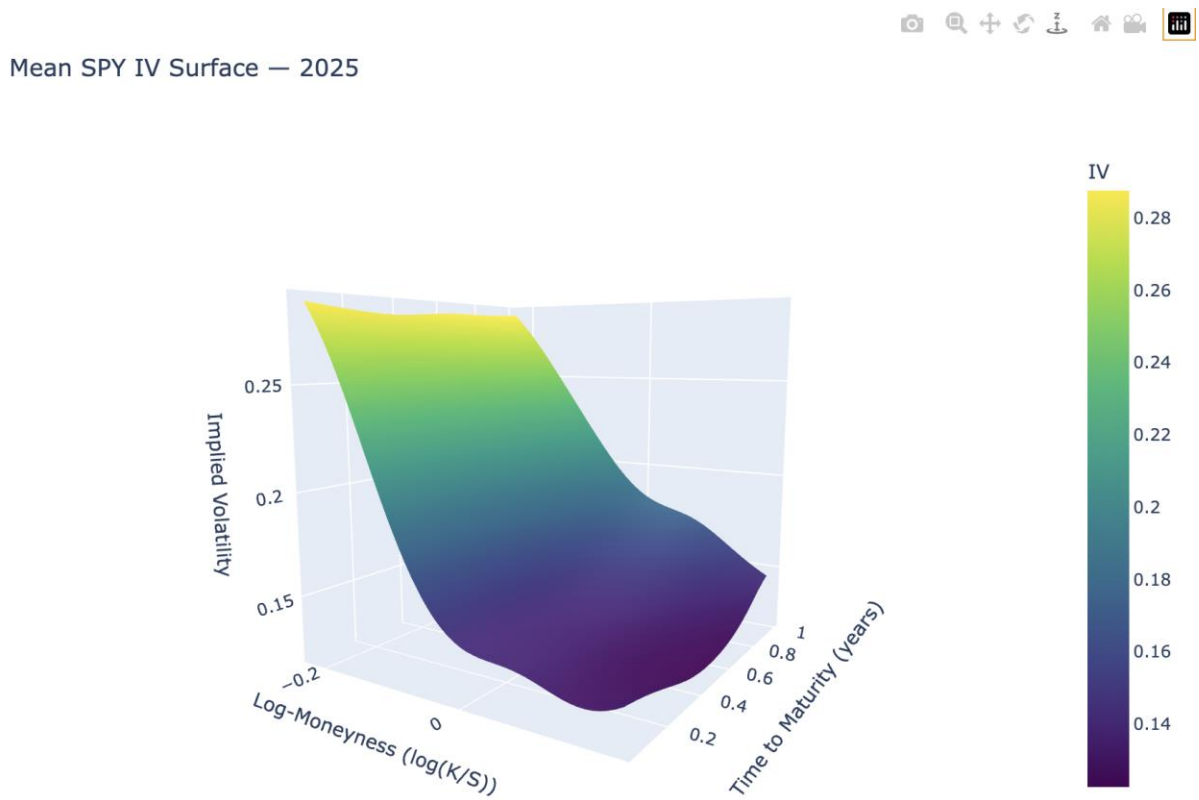
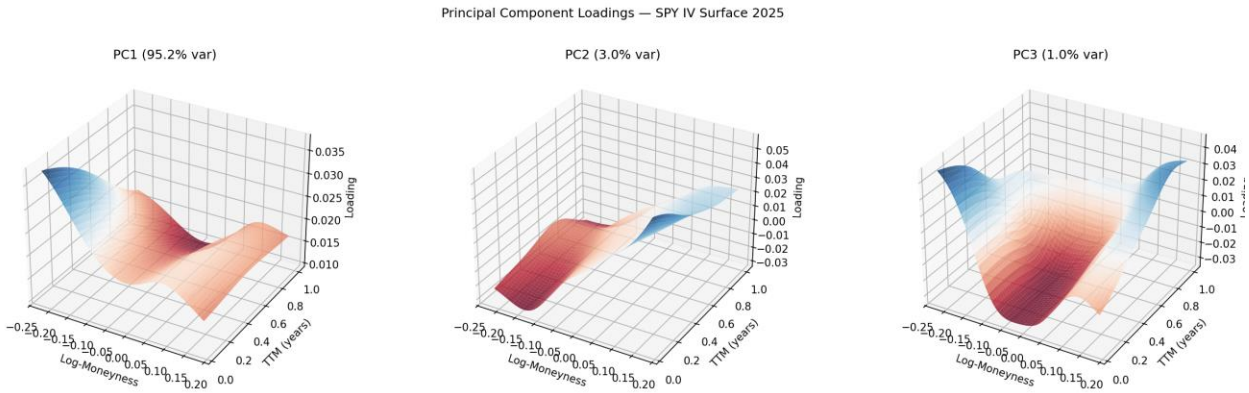
Component	Variance Explained	Cumulative
PC1 (Level)	~95%	~95%
PC2 (Slope)	~3%	~98%
PC3 (Curvature)	~1%	~99%




Principal Component Loadings — SPY IV Surface 2025
PC1 (95.2% var)
PC2 (3.0% var)
PC3 (1.0% var)
Loading
Log-Moneyness
TTM (years)

Principal Component Scores — SPY IV Surface 2025
PC1 Score Time Series (95.2% variance explained)
PC1 Score
PC2 Score Time Series (3.0% variance explained)
PC2 Score
PC3 Score Time Series (1.0% variance explained)
PC3 Score
Date